opentext™

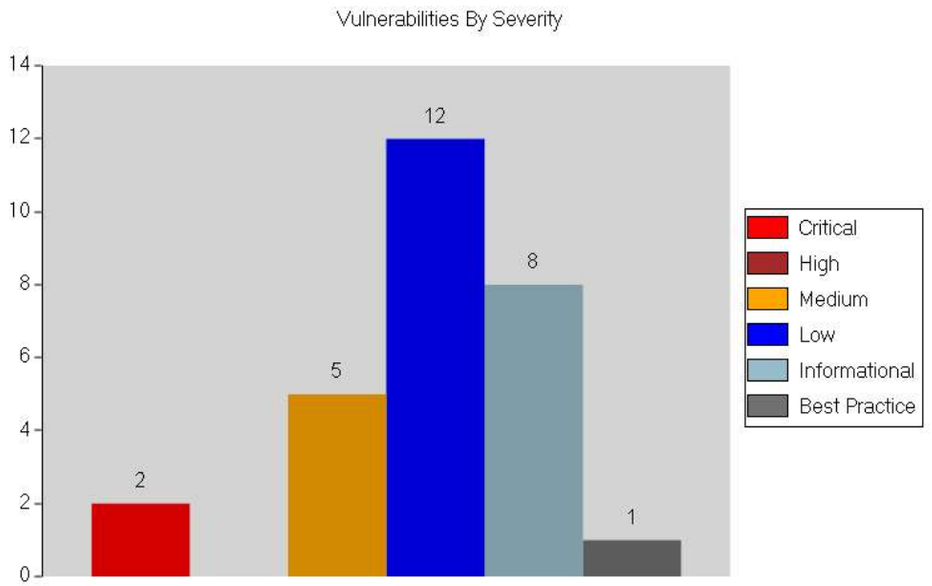
Fortify WebInspect

Vulnerability (Legacy)

Web Application Assessment Report

Scan Name:	retest_portalproyectos.reniec.gob.pe-PLPI		
Policy:	Standard	**Crawl Sessions:**	109
Scan Date:	4/15/2026 5:06:42 PM	**Vulnerabilities:**	19
Scan Version:	24.4.0.70	**Scan Duration:**	4 minutes : 46 seconds
Scan Type:	Site	**Client:**	Custom

Server: https://portalproyectos.reniec.gob.pe:443



Critical **Privacy Violation: Social Security Number**

Summary:

A critical vulnerability has been detected within your web application due to the presence of one or more Social Security Numbers. If this information is carried over to a production server, it can cause major security problems. Recommendations include not storing this information on your web application.

Implication:

Social Security Numbers are a highly sought out prize for attackers, and an item to which a large percentage of time would be dedicated in an effort to find. At a minimum, this can lead to theft of the victim's identity.

Fix:

- When sensitive data needs to be available on your web application, mask part of the data so this information is not fully disclosed.

Here are a few examples:

Social Security Numbers:
***-**-1234
123-**-****

- If presence of social security number is being reported in a JWT, please note that unless encrypted JWT tokens do not provide privacy protection. Please do include private information in JWT unless it is securely encrypted.

File Names:
- https://portalproyectos.reniec.gob.pe:443/PLPI/resources1/js/jquery.inputmask.min.js

Critical **Insecure Deployment: HTTP Request Smuggling**

Summary:

WebInspect has detected the application to be vulnerable to an HTTP Request Smuggling attack.

HTTP Request Smuggling vulnerabilities arise due to the discrepancy in parsing of non-compliant HTTP headers by the front-end and back-end servers. By supplying a request that is interpreted as being of different lengths by different servers, an attacker can poison the back-end TCP/TLS socket and prepend arbitrary data to the next request or smuggle additional requests to the back-end server without the front-end server being aware of it.

There are numerous ways in which a malicious user can accomplish an HTTP Request Smuggling attack. In this instance, an

incoming HTTP request that contains both Content-Length and Transfer-Encoding headers is interpreted differently by the front-end server and the back-end server. One honors the Content-Length header and the other the Transfer-Encoding header to determine the length of the request. This can render the application vulnerable to smuggling attacks.

The attack vector used indicates that the front-end server honors the Content-Length header and the back-end server honors the Transfer-Encoding header.

WebInspect was able to confirm this vulnerability. WebInspect performed the attack by sending two requests on the same TCP/IP connection with the server. The first request with HTTP headers

```
Content-Length: 36
Transfer-Encoding: chunked
```
and request body

```
0POST /mal formed ZZZZ/9.7Q: W
```
was crafted so that the server put part of the request body in a queue. This was prepended to the subsequent HTTP request and caused an error on the server.

Execution:

The Request Smuggling check accuracy might be impacted if the server is processing multiple requests, or if there are multiple servers behind a load balancer. The load balancer might send attack and verification requests to different backend servers and that would fail to flag the issue correctly. We recommend that you re-scan with a policy that contains only this check. Enable the traffic monitor.
Inspect the traffic monitor logs for the resulting scan if the vulnerability was flagged. Enable the column Scan.CheckID in the column view and filter on check ID 11613 to only see the traffic generated for this check.

Locate a POST request with both Content-Type and Transfer-Encoding headers and the mentioned request body. The values of these two headers is crafted so that the server put the value

```
POST /mal formed ZZZZ/9.7Q: W
```
in a queue and it is prepended to next incoming request. With this malformed start of the next HTTP request, the server is expected to respond with an error response on the normal HTTP request. This request and the response should be visible in the traffic log after the request with the attack.

Implication:

Malicious users can use the HTTP Request Smuggling vulnerability to bypass front-end security rules, access internal systems, and poison web caches. In addition, it can also be exploited to steal user information and launch cross-site scripting attacks.

Fix:

Fix this vulnerability by implementing one of the following:

Configure the front-end server to exclusively use HTTP/2 to communicate to back-end systems.
Disable back-end connection reuse.
Ensure all servers in the chain run the same webserver software with the same configuration.
Configure the back-end server to reject ambiguous HTTP requests, such as those that contain both Content-Length and Transfer-Encoding headers, and close the TCP/TLS connection.

Reference:

CWE 444 - HTTP Request Smuggling
OWASP - HTTP Request Smuggling
HTTP Desync Attacks: Request Smuggling Reborn
HTTP Request Smuggling in 2020 – New Variants, New Defenses and New Challenges

File Names:
- https://portalproyectos.reniec.gob.pe:443/PLPI/6cc675loginDatosDni

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Medium **Cookie Security: Cookie not Sent Over SSL**

Summary:

This policy states that any area of the website or web application that contains sensitive information or access to privileged functionality such as remote site administration requires that all cookies are sent via SSL during an SSL session. The URL: https://portalproyectos.reniec.gob.pe:443/PLPI has failed this policy. If a cookie is marked with the "secure" attribute, it will only be transmitted if the communications channel with the host is a secure one. Currently this means that secure cookies will only be sent to HTTPS (HTTP over SSL) servers. If secure is not specified, a cookie is considered safe to be sent in the clear over unsecured channels.

Fix:

Set the `Secure` attribute on the Cookie so that the Cookie is not sent over an unencrypted HTTP request.

Require SSL for an Authentication Cookie (IIS 7):
http://technet.microsoft.com/en-us/library/cc771633(WS.10).aspx

AnonymousIdentificationSection Class [IIS 7]:
http://msdn.microsoft.com/en-us/library/ms689482.aspx

Use the following links to remediate this issue on an Apache server:
http://search.cpan.org/~jkrasnoo/ApacheCookieEncrypted-0.03/Encrypted.pm
http://hc.apache.org/httpclient-3.x/apidocs/org/apache/commons/httpclient/class-use/Cookie.html

Set-Cookie - HTTP MDN (mozilla.org)
https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Set-Cookie#secure

Secure Cookie Attribute OWASP Foundation
https://owasp.org/www-community/controls/SecureCookieAttribute

Reference:

General Information:
The Unofficial Cookie FAQ

File Names:
- https://portalproyectos.reniec.gob.pe:443/PLPI

Medium **Privacy Violation: Shoulder Surfing**

Summary:

Basic web application security measures include "masking" all passwords to ensure they are not easily visible/recovered. This includes both passwords typed in by the user (which may be viewed by someone looking over the user's shoulder), as well as hidden passwords returned by the server (which may be viewed by looking at the HTML source code in the browser). Recommendations include requiring all password fields in your web application be masked to prevent other users from seeing this information.

Fix:

For Security Operations:
Implement a security policy that prevents display of passwords in clear text.

For Development:
Ensure that passwords entered by users on a site are not displayed in clear text. Also, do not return passwords to the user within hidden form fields.

For QA:
Ensure that passwords entered by users on a site are not displayed in clear text by logging it as a defect and notifying the appropriate developer.

File Names:
- https://portalproyectos.reniec.gob.pe:443/PLPI/loginDatosDni

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Medium **Cross-Frame Scripting**

Summary:

A Cross-Frame Scripting (XFS) vulnerability can allow an attacker to load the vulnerable application inside an HTML iframe tag on a malicious page. The attacker could use this weakness to devise a Clickjacking attack to conduct phishing, frame sniffing, social engineering or Cross-Site Request Forgery attacks.

Clickjacking
The goal of a Clickjacking attack is to deceive the victim (user) into interacting with UI elements of the attacker's choice on the target web site without their knowledge and then executing privileged functionality on the victim's behalf. To achieve this goal, the attacker must exploit the XFS vulnerability to load the attack target inside an iframe tag, hide it using Cascading Style Sheets (CSS) and overlay the phishing content on the malicious page. By placing the UI elements on the phishing page so they overlap with those on the page targeted in the attack, the attacker can ensure that the victim must interact with the UI elements on the target page not visible to the victim.

WebInspect has detected a response containing one or more forms that accept user input but is missing XFS protection.

An effective frame-busting technique was not observed while loading this page inside a frame.

Execution:

Create a test page containing an HTML iframe tag whose src attribute is set to https://portalproyectos.reniec.gob.pe:443/PLPI/resources1/js/login.js?id=20260415170655. Successful framing of the target page indicates that the application is susceptibile to XFS.

Note that WebInspect will report only one instance of this check across each host within the scope of the scan. The other visible pages on the site may, however, be vulnerable to XFS as well and therefore should be protected against it with an appropriate fix.

Implication:

A Cross-Frame Scripting weakness could allow an attacker to embed the vulnerable application inside an iframe. Exploitation of this weakness could result in:

Hijacking of user events such as keystrokes
Theft of sensitive information
Execution of privileged functionality through combination with Cross-Site Request Forgery attacks

Fix:

The Content Security Policy (CSP) frame-ancestors directive obsoletes the X-Frame-Options header. Both provide for a policy-based mitigation technique against cross-frame scripting vulnerabilities. The difference is that while the X-Frame-Options technique only checks against the top-level document's location, the CSP frame-ancestors header checks for conformity from all ancestors.

If both CSP frame-ancestors and X-Frame-Options headers are present and supported, the CSP directive will prevail. WebInspect recommends using both CSP frame-ancestors and X-Frame-Options headers as CSP is not supported by Internet Explorer and many older versions of other browsers.

In addition, developers must also use client-side frame busting JavaScript as a protection against XFS. This will enable users of older browsers that do not support the X-Frame-Options header to also be protected from Clickjacking attacks.

X-Frame-Options
Developers can use this header to instruct the browser about appropriate actions to perform if their site is included inside an iframe.Developers must set the X-Frame-Options header to one of the following permitted values:

- DENY
Deny all attempts to frame the page
- SAMEORIGIN
The page can be framed by another page only if it belongs to the same origin as the page being framed

Content-Security-Policy: frame-ancestors

Developers can use the CSP header with the frame-ancestors directive, which replaces the X-Frame-Options header, to instruct the browser about appropriate actions to perform if their site is included inside an iframe. Developers can set the frame-ancestors attribute to one of the following permitted values:

- 'none'
Equivalent to "DENY" - deny all attempts to frame the page
- 'self'
Equivalent to "SAMEORIGIN" - the page can be framed by another page only if it belongs to the same origin as the page being framed
- <scheme-source>
Developers can also specify a schema such as http: or https: that can frame the page.

Reference:

Frame Busting:
Busting Frame Busting: A Study of Clickjacking Vulnerabilities on Popular Sites
OWASP: Busting Frame Busting

OWASP:
Clickjacking

Content-Security-Policy (CSP)
CSP: frame-ancestors

Specification:
Content Security Policy Level 2
X-Frame-Options IETF Draft

Server Configuration:
IIS
Apache, nginx

HP 2012 Cyber Security Report
The X-Frame-Options header - a failure to launch

File Names:
- https://portalproyectos.reniec.gob.pe:443/PLPI/resources1/js/login.js?id=20260415170655

Medium **Insecure Deployment: OpenSSL**

Summary:

WebInspect has detected an SSL/TLS man-in-the-middle (MitM) vulnerability caused by a specially crafted SSL handshake

message. Also known as OpenSSL ChangeCipherSpec (CCS) injection vulnerability, this bug is known to manifest in the OpenSSL implementation of the secure socket layer for versions earlier than 0.9.8za, 1.0.0 before 1.0.0m and 1.0.1 before 1.0.1h. A ChangeCipherSpec message signals peers to switch to a symmetric encryption during SSL handshake process after negotiating a master key to use for symmetric encryption. However, vulnerable versions of OpenSSL allow a CCS message before the master key is negotiated resulting in a zero length master key.

Execution:

To verify the vulnerability, determine the version of the OpenSSL library deployed on the application servers. The version information can be obtained by running the command openssl version. Note that the client needs to be directly connected to the server in order to test for the vulnerability. If a proxy server is present in the environment, it is recommended to test both the proxy and the application server for the vulnerability.

Implication:

Man-in-the-middle attackers may use this vulnerability to hijack and intercept secure SSL/TLS communication by issuing an early CCS message to client and server when both are using vulnerable instance of OpenSSL. This may allow the attacker to compromise the confidentiality of sensitive session data.

Fix:

Upgrade to latest OpenSSL version.

- OpenSSL 0.9.8 SSL/TLS users should upgrade to 0.9.8za or later.
- OpenSSL 1.0.0 SSL/TLS users should upgrade to 1.0.0m or later.
- OpenSSL 1.0.1 SSL/TLS users should upgrade to 1.0.1h or later.

Additionally, users should check for the possibility of an indirect dependency on the OpenSSL library via third party software. Upgrade any such instances according to vendor recommendations.

Reference:

- https://www.openssl.org/news/secadv_20140605.txt
- http://web.nvd.nist.gov/view/vuln/detail?vulnId=CVE-2014-0224

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI

Medium **Insecure Deployment: Path Normalization Conflict**

Summary:

WebInspect has detected a Path Normalization Inconsistency Problem in a multi-layered architecture. Each URL Path may have several alternative URL representations. The goal of the normalization process is to transform a URL into a form so that all syntactical variants of the same semantic URL will be treated equally. In multi-layered architectures where the client request is routed through various intermediate components such as a load-balancer, a reverse proxy, or a web application firewall there is a risk that normalization is different at each layer. In this case, an attacker may be able to bypass allow list and deny list ACL rules or escape from context mapping. As a result, sensitive resources from the server will be unexpectedly exposed.

Example: The following configuration of NGINX forwards all requests for */publicfolder*to the Tomcat server:

```
location /publicfolder/ {
    proxy_pass http://TomcatServer:8080/;
    }
 ...
```

In the example above, it is expected that only */publicfolder*from the Tomcat server is accessible from outside.However, an attacker can send the next request: `http://targetserver/publicfolder/..;/manager/html`NGINX considers `..;` as a folder name, so the path matches the NGINX rule and is forwarded to the Tomcat server. The Tomcat server considers `..;` as a folder with a path parameter and will map the request to */publicfolder/../manager/htm*, resulting in the attacker gaining access to the manager app on the Tomcat server.

Execution:

Open a web browser and navigate to https://portalproyectos.reniec.gob.pe:443/PLPI/..;/PLPI/loginDatosDni.If the server responds with 200 OK, try to change the name of the last directory before `..;/` in the path to a non-existent directory and send the request to the server. If the server responds with 404 File Not Found error, the configuration has a Path Normalization Inconsistency.

Implication:

Path Normalization Inconsistency Problem can allow an attacker to bypass allow list, and/or deny list, ACL. Alternatively, an attacker may escape from the context mapping to gain access to sensitive resources that would otherwise be inaccessible.

Fix:

- Refer to the vendor advisory for the intermediate component and the web server. Replace the component with a version or product that will have a Path Normalization process identical to the other components and will avoid Path Normalization Inconsistency.
- Consider removing unnecessary sensitive resources from Web server.
- Add additional security protection on the Web Server such as forbidding requests from intermediate components to sensitive resources like server management console, etc. and/or add authentication to such sensitive resources.

Reference:

Breaking Parser Logic by Orange Tsai

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI/..;/PLPI/loginDatosDni

Low

Web Server Misconfiguration: Unprotected Directory

Summary:

Directory Enumeration vulnerabilities were discovered within your web application. Risks associated with an attacker discovering a directory on your application server depend upon what type of directory is discovered, and what types of files are contained within it. The primary threat, other than accessing files containing sensitive information, is that an attacker can utilize the information discovered in that directory to perform other types of attacks. Recommendations include restricting access to important directories or files by adopting a "need to know" requirement for both the document and server root, and turning off features such as Automatic Directory Listings that provide information that could be utilized by an attacker when formulating or conducting an attack.

Fix:

For Security Operations:
You should evaluate the production requirements for the found directory. If the directory is not required for production operation, then the directory and its contents should be removed or restricted by a server access control mechanism. More information about implementing access control schemes can be found in the References. Automatic directory indexing should also be disabled, if applicable.

For Development:
This problem will be resolved by the web application server administrator. In general, do not rely on 'hidden' directories within the web root that can contain sensitive resources or web applications. Assume an attacker knows about the existence of all directories and files on your web site, and protect them with proper access controls.

For QA:
This problem will be resolved by the web application server administrator.

Reference:

Implementing Basic Authentication in IIS
http://www.microsoft.com/technet/prodtechnol/WindowsServer2003/Library/IIS/abbca505-6f63-4267-aac1-1ea89d861eb4.mspx

Implementing Basic Authentication in Apache
http://httpd.apache.org/docs/howto/auth.html#intro

File Names:

- https://portalproyectos.reniec.gob.pe:443/cgi-bin/

Low

Web Server Misconfiguration: Server Error Message

Summary:

A server error response was detected. The server could be experiencing errors due to a misbehaving application, a misconfiguration, or a malicious value sent during the auditing process. While error responses in and of themselves are not dangerous, per se, the error responses give attackers insight into how the application handles error conditions. Errors that can be remotely triggered by an attacker can also potentially lead to a denial of service attack or other more severe vulnerability. Recommendations include designing and adding consistent error handling mechanisms which are capable of handling any user input to your web application, providing meaningful detail to end-users, and preventing error messages that might provide information useful to an attacker from being displayed.

Implication:

The server has issued a 500 error response. While the body content of the error page may not expose any information about the technical error, the fact that an error occurred is confirmed by the 500 status code. Knowing whether certain inputs trigger a server error can aid or inform an attacker of potential vulnerabilities.

Fix:

For Security Operations:

Server error messages, such as "File Protected Against Access", often reveal more information than intended. For instance, an attacker who receives this message can be relatively certain that file exists, which might give him the information he needs to pursue other leads, or to perform an actual exploit. The following recommendations will help to ensure that a potential attacker is not deriving valuable information from any server error message that is presented.

- Uniform Error Codes: Ensure that you are not inadvertently supplying information to an attacker via the use of inconsistent or "conflicting" error messages. For instance, don't reveal unintended information by utilizing error messages such as Access Denied, which will also let an attacker know that the file he seeks actually exists. Have consistent terminology for files and folders that do exist, do not exist, and which have read access denied.
- Informational Error Messages: Ensure that error messages do not reveal too much information. Complete or partial paths, variable and file names, row and column names in tables, and specific database errors should never be revealed to the end user. Remember, an attacker will gather as much information as possible, and then add pieces of seemingly innocuous information together to craft a method of attack.
- Proper Error Handling: Utilize generic error pages and error handling logic to inform end users of potential problems. Do not provide system information or other data that could be utilized by an attacker when orchestrating an attack.

Removing Detailed Error Messages

Find instructions for turning off detailed error messaging in IIS at this link:

http://support.microsoft.com/kb/294807

For Development:

From a development perspective, the best method of preventing problems from arising from server error messages is to adopt secure programming techniques that prevent problems that might arise from an attacker discovering too much information about the architecture and design of your web application. The following recommendations can be used as a basis for that.

- Stringently define the data type (for instance, a string, an alphanumeric character, etc) that the application will accept.
- Use what is good instead of what is bad. Validate input for improper characters.
- Do not display error messages to the end user that provide information (such as table names) that could be utilized in orchestrating an attack.
- Define the allowed set of characters. For instance, if a field is to receive a number, only let that field accept numbers.
- Define the maximum and minimum data lengths for what the application will accept.
- Specify acceptable numeric ranges for input.

For QA:

The best course of action for QA associates to take is to ensure that the error handling scheme is consistent. Do you receive a different type of error for a file that does not exist as opposed to a file that does? Are phrases like "Permission Denied" utilized which could reveal the existence of a file to an attacker? Inconsistent methods of dealing with errors gives an attacker a very powerful way of gathering information about your web application.

Reference:

Apache:
Security Tips for Server Configuration
Protecting Confidential Documents at Your Site
Securing Apache - Access Control

Microsoft:
How to set required NTFS permissions and user rights for an IIS 5.0 Web server
Default permissions and user rights for IIS 6.0
Description of Microsoft Internet Information Services (IIS) 5.0 and 6.0 status codes

File Names:

- https://portalproyectos.reniec.gob.pe:443/usr/share/zoneinfo/GMT%3ffakestp.html
- https://portalproyectos.reniec.gob.pe:443/cgi-bin/README.txt
- https://portalproyectos.reniec.gob.pe:443/PLPI/GET
- https://portalproyectos.reniec.gob.pe:443/PLPI/usr/share/zoneinfo/GMT%3ffakestp.html
- https://portalproyectos.reniec.gob.pe:443/GET
- https://portalproyectos.reniec.gob.pe:443/<script>alert('TRACK');</script>

Low **Cache Management: Insecure Policy**

Summary:

WebInspect has detected a potentially unsafe cache control policy for secure content. While content transmitted over an SSL/TLS channel is expected to guarantee confidentiality, administrators must ensure that caching of sensitive content is disabled unless absolutely needed. The misconception that secure content caching is disabled by default by user-agents could cause the application to fail the organization's cache policy. An unsafe specification such as Cache-Control: public instructs the browser to persistently cache the content on the hard drive. Cache-Control with no-store in the value must be set to prevent browsers from persisting content. Browsers and intermediate proxies will still persists with the no-cache directive. However, they will revalidate the content with the server before serving content from cache. The private directive prevents intermediate proxies from caching content. Missing Cache-Control policy header results in browsers caching content regardless of whether it is served over HTTP or HTTPS.

- Cache-Control: no-store

Execution:

Send a request to https://portalproyectos.reniec.gob.pe:443/PLPI and inspect the Cache-Control header value.

Implication:

Insecure caching policies could lead to content spoofing or information theft.

SSL provides secure encrypted channel to transfer information from source to user. The information served over SSL is considered sensitive and trusted to be only available to the requestor. However, caching this content on disk in temporary internet files or on an intermediate proxy server can compromise that trust by exposing it to anyone who has access to the temporary storage or proxy cache. Content served over SSL should have cache disabled.

Fix:

Set the Cache-Control directive to no-store.

no-cache
For sensitive resources requiring user authentication, servers can send the no-cache directive to prevent caches from serving a cached response without first requiring the user-agent to validate the user identity. This directive can be specified with or without field names. When no field names are included, this directive applies to the entire request or response.
When one or more field names are specified in the no-cache directive, the response is cached but the specified fields must be excluded. If the response must include the specified field, then the cache must ensure that the request triggers a revalidation with the origin server.
Example: Cache-Control: no-cache="Set-Cookie"
This directive can be used to prevent sensitive information leakage by requiring the server to confirm the user identity before serving the protected information. While content is not served using the browser's history or back button, a user with disk access can retrieve the content.

no-store
To completely disable the caching of requests or responses, the server must specify the no-store directive in the Cache-Control header. This directive applies to the entire request and response regardless of whether the directive is sent in the request or the response.

Reference:

Server Configuration:
IIS
http://support.microsoft.com/kb/247404
Apache
https://httpd.apache.org/docs/2.4/caching.html

RFC 9111 Section 5.2 Cache Control
https://datatracker.ietf.org/doc/html/rfc9111#section-5.2

Browser Cache FAQ
https://www.owasp.org/index.php/OWASP_Application_Security_FAQ#Browser_Cache

Caching Tutorial
https://www.mnot.net/cache_docs/

File Names:
- https://portalproyectos.reniec.gob.pe:443/PLPI

Low **Insecure Transport: HSTS not Set**

Summary:

Http Strict Transport Security (HSTS) policy enables web applications to enforce web browsers to restrict communication with the server over an encrypted SSL/TLS connection for a set period. Policy is declared via special Strict Transport Security response header. Encrypted connection protects sensitive user and session data from attackers eavesdropping on network connection.
Consider following attack scenarios:

- Users often omit the URI scheme i.e. https:// when typing a URL in location bar to access a website. Also third party websites can link to the site using the "http" scheme instead of ""https". This could result in an initial connection to a HTTPS-enabled site over an unencrypted channel. An eavesdropping attacker can hijack this unencrypted connection and replace the intended use of HTTPS protocol with HTTP in an attack known as SSLStrip, granting unauthorized access to

all subsequent traffic.
- Websites often transfer non-sensitive resources such as help documents over an unencrypted HTTP connection. Any cookies without a secure flag are sent along with such requests potentially disclosing sensitive user and session data to eavesdropper.
- Man-in-the-Middle attacks that exploit user tendencies to override invalid certification warnings, e.g. SSLSniff.

For web sites configured with an accurate HSTS policy, browsers automatically upgrade any HTTP connections to HTTPS. Furthermore, browsers prevent users from overriding any host certificate warnings. HSTS offers an effective defense against above attack scenarios.

Execution:

Access location https://portalproyectos.reniec.gob.pe:443/ and notice the absence of the Strict Transport Security header in the HTTP response.

Implication:

A successful MiTM attack such as SSLStrip or SSLsniff can lead to the compromise of sensitive user data such as financial information, Social Security Number, personal information etc. as well as grant unauthorized access to user accounts enabling attackers to perform privileged actions on client's behalf.

Fix:

Configure the web application under test to include Strict Transport Security header in every response generated by an HTTPS -enabled site. Any HTTP version of site on the same domain should permanently redirect to the secure encrypted site. Header should not be added to HTTP response as browsers will ignore it.

It is important to note that this header does not prevent from above mentioned attack scenarios during the very first connection to the site or any connections established after the set period has expired. To prevent such a scenario, the site must be added to the pre-loaded HSTS hosts list embedded in both Google Chrome and Mozilla Firefox browsers.

Reference:

http://tools.ietf.org/html/rfc6797

File Names:
- https://portalproyectos.reniec.gob.pe:443/

Low **HTML5: Misconfigured Content Security Policy**

Summary:

Content Security Policy (CSP) is a declarative security header that allows developers to dictate which domains the site is allowed to load contents from or initiate connection to when rendered in the web browser. It provides an additional layer of security from critical vulnerabilities like cross site scripting, clickjacking, cross origin access etc. on top of input validation and having an allow list in code. Improperly configured header however, fails to provide this additional layer of security. The policy is defined with the help of fifteen directives including eight that control resource access namely -

- script-src
- img-src
- object-src
- style_src
- font-src
- media-src
- frame-src
- connect-src

Each of these takes a source list as a value specifying domains the site is allowed to access for feature covered by that directive. Developers may use wildcard * to indicate all or part of the source. None of the directives are mandatory. Browsers will either allow all sources for unlisted directive or will derive its value from optional default-src directive. Furthermore, the specification for this header has evolved over time. It was implemented as X-Content-Security-Policy in Firefox until version 23 and in IE until version 10, and was implemented as X-Webkit-CSP in Chrome until version 25. Both of the names are deprecated in favor of the now standard name Content Security Policy. Given number of directives, two deprecated alternate names, and the way multiple occurrences of same header and repeat directive in single header is treated there is a high propensity that developer might misconfigure this header.

Consider following misconfiguration scenarios:

- It can be an overly permissive policy if default-src is not set or set to wildcard and/or other directives are set to wildcard.
- Multiple instances of this header are allowed in same response. A development team and security team might both set the header but with either different policies or one may use now deprecated X-Content-Security-Poilcy or X-Webkit-CSP headers. Deprecated versions are ignored if standard Content-Security-Policy header is present.
- If a directive is repeated within the same instance of the header, all subsequent occurrences are ignored.

Following misconfigurations were detected at https://portalproyectos.reniec.gob.pe:443/PLPI:

Overly Permissive Content Security Policy.

- *Content-Security-Policy:default-src,script-src,object-src,font-src,style-src,img-src,frame-src,connect-src,media-src* are set to wildcard.

Execution:

Access link https://portalproyectos.reniec.gob.pe:443/PLPI through proxy and notice the misconfigured content security policy header in response headers.

Implication:

Content security policy header provides additional layer of security for site from client side attacks such as cross-site scripting. In absence of this header, an attacker can exploit client side vulnerabilities such as cross-site scripting, clickjacking and cross-site request forgery.

Fix:

Remove wildcard values for source-list to limit scope of cross origin access from site. Ensure canonical *Content Security Policy* name is used to specify policy. X-Content-Security-Policy and X-Webkit-CSP names are deprecated. Any references to these headers are only useful in case support for earlier browsers is desired. Presence of these headers in addition to Content-Security-Policy is said to cause unexpected behaviors on certain versions of browsers.
Content-Security-Policy version 2 standard support:

- Edge: Edge 15 - 18 supported with a nonce bug. Version 75 onwards fully supported.
- Chrome: Chrome 36-38 are missing the plugin-types, child-src, frame-ancestors, base-uri, and form-action directives. Chrome 39 is missing the plugin-types, child-src, base-uri, and form-action directives. Chrome 40 onwards fully supported.
- Firefox; Firefox 31-34 is missing the plugin-types, child-src, frame-ancestors, base-uri, and form-action directives. Firefox 35 is missing the plugin-types, child-src, frame-ancestors, and form-action directives. Firefox 36-44 is missing the plugin-types and child-src directives. Firefox 45+ is missing the plugin-types directive.

Furthermore, the report-uri directive can be configured to receive reports of attempts to violate the policy. These reports can be used as an early indication of security issues in the site as well as to optimize the policy.

Reference:

http://www.w3.org/TR/CSP/
https://owasp.org/www-community/controls/Content_Security_Policy
https://developer.mozilla.org/en-US/docs/Web/Security/CSP/Introducing_Content_Security_Policy
https://content-security-policy.com

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI

Low **Cookie Security: Missing SameSite Attribute**

Summary:

The SameSite attribute protects cookies from Cross-Site Request Forgery (CSRF) attacks. The browser automatically appends cookies to every HTTP request made to the site that sets the cookie. Cookies might store sensitive data like session ID and authorization token or site data that is shared between different requests to the same site during a session. An attacker can perform an impersonation attack by generating a request to the authenticated site from a third-party site page loaded on the client machine because the browser automatically appended the cookie to the request.
The SameSite attribute on a cookie allows sites to control that behaviour and prevents browsers from appending the cookie to request if the request is generated from a third-party site page load. The SameSite attribute can have the following three values:

- Strict: When set to Strict, cookies are only sent along with requests upon top level navigation .
- Lax: When set to Lax, cookies are sent with top level navigation from the same host as well as GET requests originated to the host from third-party sites (for example, in iframe, link, href, and so on and the form tag with GET method only).
- None: Cookies are sent in all requests made to the site within the path and domain scope set for the cookie. Requests generated due to form submissions using the POST method are also allowed to send cookies with request.

Please note that cookies that have the SameSite attribute with the value of None must be set with the Secure attribute otherwise the browser rejects the cookies. Additionally, a few specific browser versions reject the SameSite cookie with the None value for example, Chrome versions 51 to 66, versions of the UC Browser on Android prior to version 12.13.2, versions of Safari and embedded browsers on macOS 10.14, and all browsers on iOS 12 reject cookies set with SameSite=None. A suggested workaround for this issue is to set an alternate cookie with a prefix or suffix such as *Legacy*appended to cookiename . Sites can look for this legacy cookie if it does not find a cookie that was set with SameSite=None.

Execution:

Inspect the highlighted cookie value in the HTTP response in the vulnerable session. The cookie is missing the SameSite attribute.

Implication:

Sites that set cookies without the SameSite attribute have an increased risk of CSRF attacks. Using CSRF, an attacker can impersonate a valid user and gain unauthorized access to application functionality. Furthermore, the recent versions of browsers might reject the cookie that is not set with the SameSite attribute.

Fix:

Add the SameSite attribute to all cookies. Starting February 2020, the Chrome browser made SameSite a mandatory attribute for all cookies. Any cookie without the SameSite attribute is either rejected or treated with the default behaviour, which is the same as setting the attribute value to Lax. Therefore, any cookie that must be sent regardless the origination of the request (for example, analytics cookies) should have the SameSite attribute value of none.
Furthermore, we recommend that developers continue to add traditional CSRF mitigations to the site along with SameSite attribute. Many site users might still use older browser versions to access the site. Older browser versions do not understand the SameSite attribute.

Reference:

https://tools.ietf.org/html/draft-ietf-httpbis-rfc6265bis-05
https://www.chromium.org/updates/same-site/incompatible-clients
https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Set-Cookie/SameSite

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI

Low

HTML5: Deprecated Header

Summary:

`X-XSS-Protection` header is enabled. `X-XSS-Protection` refers to a header used by developers to control the browser's behavior and offers protection against cross-site scripting (XSS). The header was designed to filter pages to remove unsafe components or stop loading when cross-site scripting was detected. Internet Explorer 8 introduced the `X-XSS-Protection` header and it was also supported in Edge version 12-16, Chrome version 4-77, Opera version 15-64, and Safari 5-15.3. Modern browsers no longer maintain, support, or enhance implementation of this header.

Having this setting enabled does not necessarily result in a vulnerability, but in some cases such filters are leveraged to perform XSS attacks against users that would otherwise not be possible. The XSS filters in the browser use regular expressions to alter the response. For example, a regex to detect `<script>` tag would alter the tag in the response to `<sc#ipt>`. It would stop the current XSS attack but also have unintended consequences. An attacker could use this knowledge to disable legitimate scripts or client-side scripts containing security features by adding a malicious parameter. By setting the `X-XSS-Protection` to `"1; mode=block"` stops the entire page from loading but it is vulnerable to side-channel attacks [3].

This header can be set explicitly in an application server configuration or in application code. All places should be checked for insecure setting of this header. This check is reported once per host by default and can be made to report all instances by setting check input `"X-XSS-Protection_Aggressive Reporting"`

Execution:

Click the response tab of the highlighted request. You will notice the occurrence of `X-XSS-Protection` header highlighted and value is set to 1.

Implication:

Setting `X-XSS-Protection` header is ineffective for browsers that no longer support this header. Enabling `X-XSS-Protection` header might increase the risk of cross-site scripting attacks against the application if a user accesses it from an older browser that supported this feature.

Fix:

Remove the response header `X-XSS-Protection`.
OpenText recommends using `Content Security Policy (CSP)` to protect against XSS attacks.

Reference:

1 X-XSS-Protection - HTTP MDN
https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-XSS-Protection

2 IE8 XSS Filters
https://media.blackhat.com/bh-eu-10/presentations/Lindsay_Nava/BlackHat-EU-2010-Lindsay-Nava-IE8-XSS-Filters-slides.pdf

3 Abusing Chrome's XSS auditor to steal tokens PortSwigger Research
https://portswigger.net/research/abusing-chromes-xss-auditor-to-steal-tokens

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI

Informational

Hidden Field

Summary:

While preventing display of information on the web page itself, the information submitted via hidden form fields is easily accessible, and could give an attacker valuable information that would prove helpful in escalating his attack methodology. Recommendations include not relying on hidden form fields as a security solution for any area of the web application that contains sensitive information or access to privileged functionality such as remote site administration functionality.

Execution:

Any attacker could bypass a hidden form field security solution by viewing the source code of that particular page.

Implication:

The greatest danger from exploitation of a hidden form field design vulnerability is that the attacker will gain information that will help in orchestrating a far more dangerous attack.

Fix:

Do not rely on hidden form fields as a method of passing sensitive information or maintaining session state. One workable bypass is to encrypt the hidden values in a form, and then decrypt them when that information is to be utilized by a database operation or a script. From a security standpoint, the best method of temporarily storing information required by different forms is to utilize a session cookie.

Whether hidden or not, if your site utilizes values submitted via a form to construct database queries, do not make the assumption that the data is non-malicious. Instead, utilize the following recommendations to sanitize user supplied input.

- Stringently define the data type (for instance, a string, an alphanumeric character, etc) that the application will accept.
- Use what is good instead of what is bad.
- Validate input for improper characters.
- Do not display error messages to the end user that provide information (such as table names) that could be utilized in orchestrating an attack.
- Define the allowed set of characters. For instance, if a field is to receive a number, only let that field accept numbers.
- Define the maximum and minimum data lengths for what the application will accept.
- Specify acceptable numeric ranges for input.

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI
- https://portalproyectos.reniec.gob.pe:443/PLPI/loginDatosDni
- https://portalproyectos.reniec.gob.pe:443/PLPI/login?lg=1
- https://portalproyectos.reniec.gob.pe:443/PLPI/sendLogin

Informational **Web Server Misconfiguration: OPTIONS HTTP Method**

Summary:

The server supports the OPTIONS HTTP method. The OPTIONS method is used to determine what other methods the server supports for a given URI/resource.

Reference:

RFC 2616 Section 9: HTTP Methods:
http://www.w3.org/Protocols/rfc2616/rfc2616-sec9.html

Apache:
Apache HTTP Server Version 2.0
Apache HTTP Server Version 1.3

Microsoft:
UrlScan Security Tool
How to configure the URLScan Tool
Setting Application Mappings in IIS 6.0

File Names:

- https://portalproyectos.reniec.gob.pe:443/

Informational **HTML5: Missing Content Security Policy**

Summary:

Content Security Policy (CSP) is an HTTP response security header that developers and security architects can leverage to create an allow list of domains from which the site is allowed to load resources. This header provides an in-depth security protection from critical vulnerabilities such as cross-site scripting and clickjacking. Additionally, CSP restricts execution of inline JavaScript, dynamic JavaScript code evaluation from strings, and framing of the site from external domains. While CSP is not a replacement for input validation, it can help to significantly reduce the risk of XSS from unknown weaknesses. The CSP frame-ancestors directive is equivalent to X-Frame-Options and restricts the domain that are allowed to frame the site's content.

Execution:

Access link https://portalproyectos.reniec.gob.pe:443/GET through a proxy and notice the missing CSP header in the response. By default, WebInspect flags only one instance of this vulnerability per host because it is typical to set this header at the host level in a server configuration.

Perform the following steps to flag all instances of this issue:

- Create a new policy with the selection of checks that you want to include in a rescan. We recommend using the Blank or Passive policy as a base.
- Select this check and uncheck the "FlagAtHost" check input from standard description.
- Save the policy.
- Rescan with this new custom policy.

Implication:

Security architects and developers can leverage CSP to significantly reduce the risk of XSS and clickjacking attacks. CSP headers can restrict leakage of information to external domains by restricting which domains the site is allowed to load contents from when rendered in browser .

Fix:

Define a CSP policy suitable for your site. The policy can be set either with an HTTP response header or <meta /> tag.

For example:
```
Content-Security-Policy: default-src https://example.net; child-src 'none';
```
Or
```
<meta http-equiv="Content-Security-Policy" content="default-src https://cdn.example.net;
child-src 'none'; object-src 'none'">
```

Content-Security-Policy 2 is the recommended standard. Content-Security-Policy 3 is in draft. The following is a snapshot of modern browser support for the CSP header:

- Edge: Versions 15-18; supported with a nonce bug. Version 75 and later; fully supported.
- Chrome: Versions 36-38; missing the plugin-types, child-src, frame-ancestors, base-uri, and form-action directives. Version 39; missing the plugin-types, child-src, base-uri, and form-action directives. Version 40 and later; fully supported.
- Firefox: Versions 31-34; missing the plugin-types, child-src, frame-ancestors, base-uri, and form-action directives. Version 35; missing the plugin-types, child-src, frame-ancestors, and form-action directives. Versions 36-44; missing the plugin-types and child-src directives. Version 45 and later; missing the plugin-types directive.

Furthermore, the report-uri directive can be configured to receive reports of attempts to violate the policy. These reports can be used as an early indication of security issues in the site as well as to optimize the policy.

Reference:

Content Security Policy Level 3
OWASP Content Security Policy
MDN web docs
Content Security Policy (CSP) Quick Reference Guide

File Names:
- https://portalproyectos.reniec.gob.pe:443/GET

Informational **HLI: Detected Libraries**

Summary:

Hacker Level Insights provides developers and security professionals with more context relating to the overall security posture of their application. The **Bootstrap** version **3.3.4** was detected to be in use by during this scan. While these findings do not necessarily represent a security vulnerability, it is important to note that attackers commonly perform reconnaissance of their target in an attempt to identify known weaknesses or patterns. Knowing what the hacker can see provides context which can help teams better secure their applications.
.cvc-grid {width:100%;}.cvc-box {border-radius: 15px; border: 2px solid #1a75ff; width: 100%; padding: 10px;}.cvc-cat {position: relative; display: inline-block; border-bottom: 1px dotted black;}.cvc-cat .nvd-tooltip {visibility: hidden; width: 400px; background-color: #3b96ff; color: #fff; text-align: left; border-radius: 6px; padding: 5px; position: absolute; z-index: 1;}.cvc-cat:hover .nvd-tooltip {visibility: visible;}.score-box{color:#000!important;background-color:#f1f1f1!important;width:75%;border-radius:8px}.score-box div {border-radius:8px}.score-box div div{padding-left: 10px; font-weight: bold;}.score-box > div:after,.score-box > div:before {content:""; display:table;clear:both}.sb-good{color:#fff!important;background-color:#87e547!important}.sb-warn{color:#fff!important;background-color:#fbc02d!important}.sb-bad {color:#fff!important;background-color:#ff8686!important}.nvd-tooltip:before{content:"The National Vulnerability Database (NVD) helps Fortify WebInspect HLI analysis to find information for CVEs including a brief summary by using Common Platform Enumeration(CPE) queries. The database is maintained by the National Institute of Standards and Technology (NIST)."}

Known NVD records for: cpe:2.3:a:getbootstrap:bootstrap:3.3.4

VULNERABILITY:	FOUND VERSION:	FIXED VERSION:	SEVERITY:
CVE-2018-14040	3.3.4	3.4.0	Medium

In Bootstrap before 4.1.2, XSS is possible in the collapse data-parent attribute.

VULNERABILITY:	FOUND VERSION:	FIXED VERSION:	SEVERITY:
CVE-2018-14042	3.3.4	3.4.0	Medium

In Bootstrap before 4.1.2, XSS is possible in the data-container property of tooltip.

VULNERABILITY:	FOUND VERSION:	FIXED VERSION:	SEVERITY:
CVE-2016-10735	3.3.4	3.4.0	Medium

In Bootstrap 3.x before 3.4.0 and 4.x-beta before 4.0.0-beta.2, XSS is possible in the data-target attribute, a different vulnerability than CVE-2018-14041.

VULNERABILITY:	FOUND VERSION:	FIXED VERSION:	SEVERITY:
CVE-2018-20676	3.3.4	3.4.0	Medium

In Bootstrap before 3.4.0, XSS is possible in the tooltip data-viewport attribute.

VULNERABILITY:	FOUND VERSION:	FIXED VERSION:	SEVERITY:
CVE-2018-20677	3.3.4	3.4.0	Medium

In Bootstrap before 3.4.0, XSS is possible in the affix configuration target property.

VULNERABILITY:	FOUND VERSION:	FIXED VERSION:	SEVERITY:
CVE-2019-8331	3.3.4	3.4.1	Medium

In Bootstrap before 3.4.1 and 4.3.x before 4.3.1, XSS is possible in the tooltip or popover data-template attribute.

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI
- https://portalproyectos.reniec.gob.pe:443/PLPI

Best Practice **Compliance Failure: Missing Privacy Policy**

Summary:

A privacy policy was not supplied by the web application within the scope of this audit. Many legislative initiatives require that organizations place a publicly accessible document within their web application that defines their website's privacy policy. As a general rule, these privacy policies must detail what information an organization collects, the purpose for collecting it, potential avenues of disclosure, and methods for addressing potential grievances.
Various laws governing privacy policies include the Gramm-Leach-Bliley Act, Health Insurance Portability and Accountability Act (HIPAA), the California Online Privacy Protection Act of 2003, European Union's Data Protection Directive and others.

Execution:

All of the web pages accessible within the scope of the scan are sampled for textual content that often constitutes a privacy policy statement. A violation is reported upon completion of the web application crawl without a successful match against any of the web pages.

Note that the privacy policy of your application could be located on another host or within a section of the site that was not configured as part of the scan. To validate, please try to access the privacy policy of your website and check to see if it was part of the scan.

Implication:

Most privacy laws are created to protect residents who are users of the website. Hence, organizations from any part of the world must adhere to these laws if they cater to customers residing in these geographical areas. Failing to do so could result in a lawsuit by the corresponding government against the organization.

Fix:

Declare a comprehensive privacy policy for the website, and ensure that it is accessible from every page that seeks personal information from users. To verify the fix, rescan the site in order to discover and audit the newly added resources.

Descriptions:
Any standard web application privacy policy should include the following components:

- A description of the intended purpose for collecting the data.
- A description of the use of the data.
- Methods for limiting the use and disclosure of the information.
- A list of the types of third parties to whom the information might be disclosed.
- Contact information for inquires and complaints.

Reference:

California Online Privacy Protection Act
http://oag.ca.gov/privacy/COPPA

National Conference of State Legislation
http://www.ncsl.org/issues-research/telecom/state-laws-related-to-internet-privacy.aspx

Gramm-Leach-Bliley Act
http://www.gpo.gov/fdsys/pkg/PLAW-106publ102/pdf/PLAW-106publ102.pdf

Health Insurance Portability and Accountability Act of 1996
https://www.cms.gov/Regulations-and-Guidance/HIPAA-Administrative-Simplification/HIPAAGenInfo/downloads/HIPAALaw.pdf

Health Insurance Portability and Accountability Act of 1996
http://ec.europa.eu/justice/policies/privacy/docs/guide/guide-ukingdom_en.pdf

File Names:

- https://portalproyectos.reniec.gob.pe:443/PLPI